

Abstract

The Nutrition Tracker is a Java desktop application that helps users log daily food intake and monitor macro- and micro-nutrient goals. The system provides two roles: User and Admin. Users can add custom foods (per 100g), log consumption, view nutrient totals for a selected date, and browse history. Admins can manage accounts, review any user's logs by date, and leave notes/instructions.

The application is built with Java (Swing) for the UI, JDBC for persistence, and MySQL for the database. Core modules include authentication, food catalog, food logging, nutrient summaries, history views, user management, and admin notes. The system reduces manual calculation errors, centralizes records, and offers clear daily/period summaries aligned with goals.

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1. Introduction

Recording diet data with paper or spreadsheets can be inefficient and prone to error. For users, it is common to encounter dispersed entries, repetitive calculations, and a limited ability to cope with and review their progress efficiently. To help solve these issues, the Nutrition Tracker provides a centralized platform where users can input their daily intake in grams, keep a bespoke catalogue of food, including nutrition information per 100 grams, and find their actual intake of nutrients quickly. Comparing achievement of targets is easy with automatic daily totals and complete breakdowns of the intake.

The system outlines administrative oversight features including administrative (usage_id) user control, daily logs sorting for administrators, and ability to leave guidance. Separation of the node for data access layer (DAO using JDBC) from user interface gives the system greater maintainability and dependability. Additionally, the database (schema) structure promotes query consistency and as well as integrity of data through relationships and indices - for example (user_id, date) for food logs.

2. Problem Statement

Conventional diet tracking techniques frequently depend on spreadsheets or paper-based records, which can result in several problems. The process of manually converting nutritional values from per-100-gram data into the actual consumed amount is extremely error-prone, and data is frequently inconsistent, with duplicate or missing entries. This lowers the motivation of users who find it difficult to maintain trustworthy logs and compromises the accuracy of dietary monitoring.

Moreover, since these methods do not illustrate continuous or cumulative progress towards daily or long-term dietary objectives, it might be hard for users to change their diets despite using these methods. The absence of structure that allows coaches, administrators, or dietitians to systematically track and coach the user presents a joint problem. The Nutrition Tracker offers these solutions through the centralization of food records, automatic calculations, and building your data in a way that is easily understood, to allow administrators to track users' progress and provide timely support.

3. Objectives

Primary Objective: Deliver a robust, user-friendly application to track daily intake and compare it against goals.

Specific Objectives:

1. Authentication & Roles — Secure login with roles (user/admin).
2. Food Catalog — Per-user foods with nutrients per 100g; CRUD operations.
3. Food Logging — Record grams consumed per day; compute nutrient totals.
4. Daily Summary — Itemized table + totals for a selected date.
5. History View — Date picker with past logs and totals.
6. Admin Oversight — Manage accounts, view user logs by date, add coach notes.
7. Data Integrity — Use foreign keys and indices for consistency/performance.
8. Scalability — Design for later features (payments, mobile client, cloud sync, analytics).

4. Diagrams

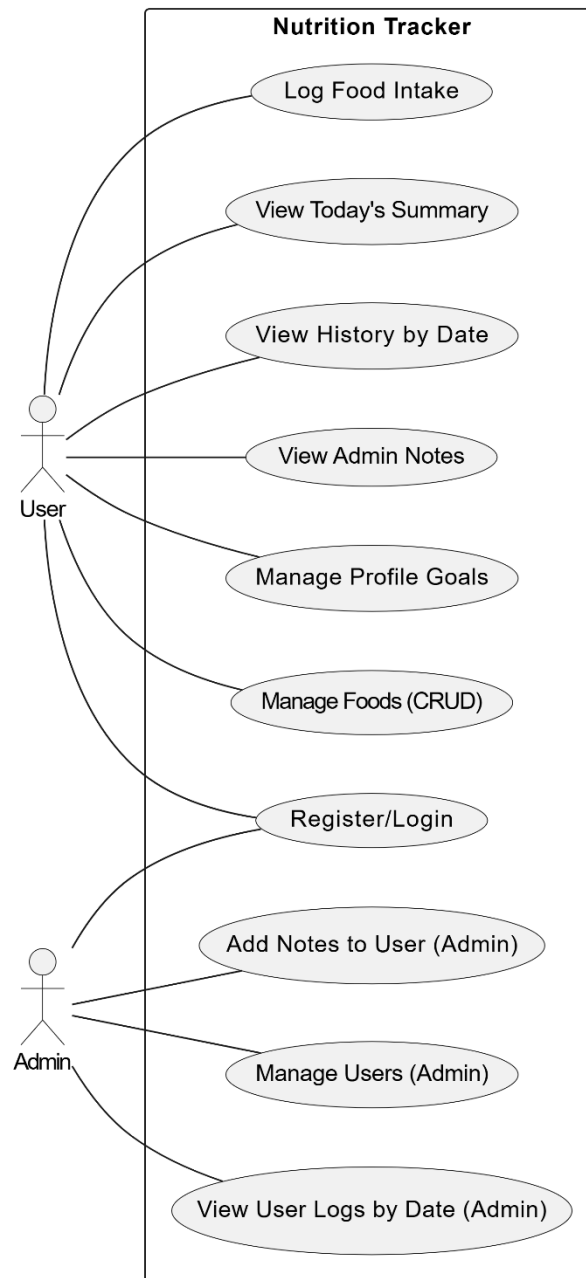


Fig: Use Case Diagram of
Nutrition Tracker

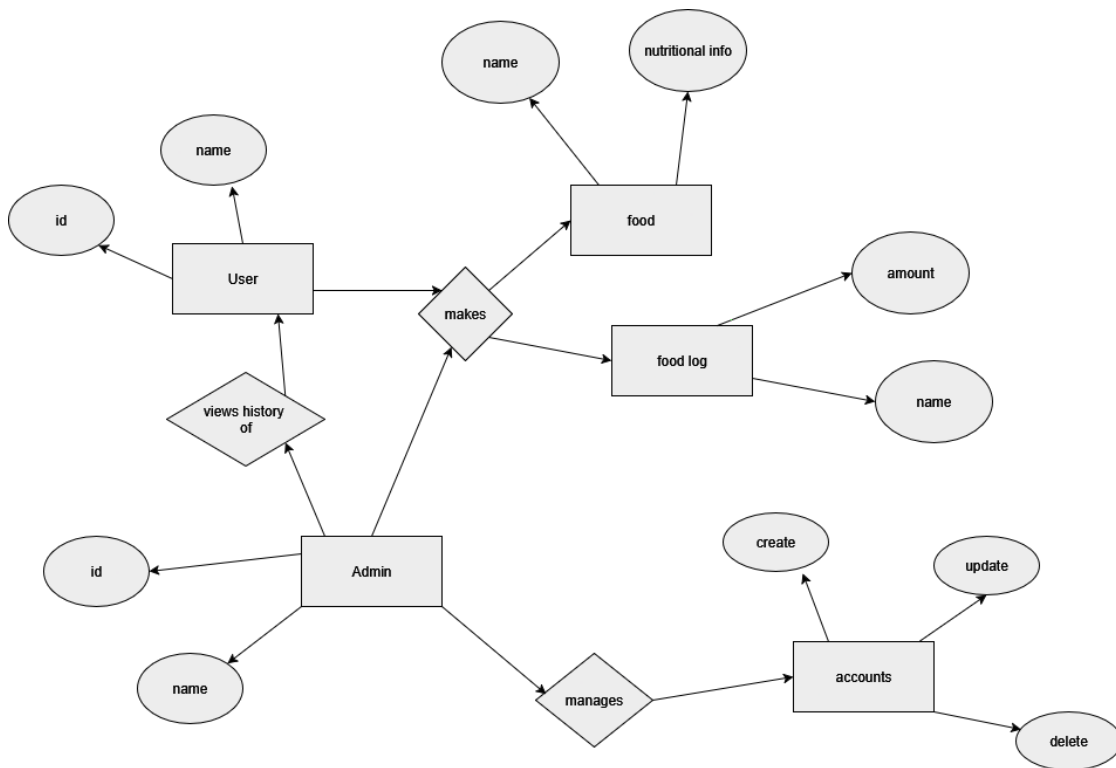
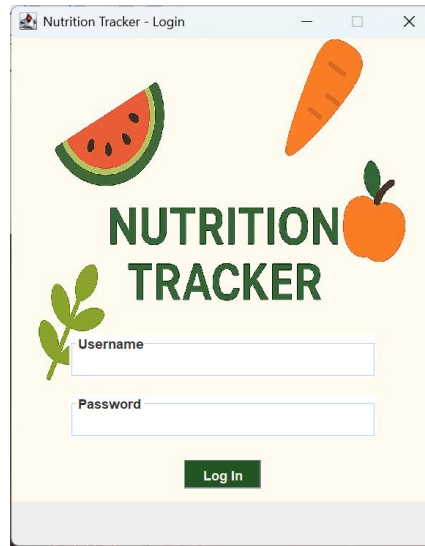


Fig: ER Diagram of Nutrition Tracker

5. Screenshot of Application



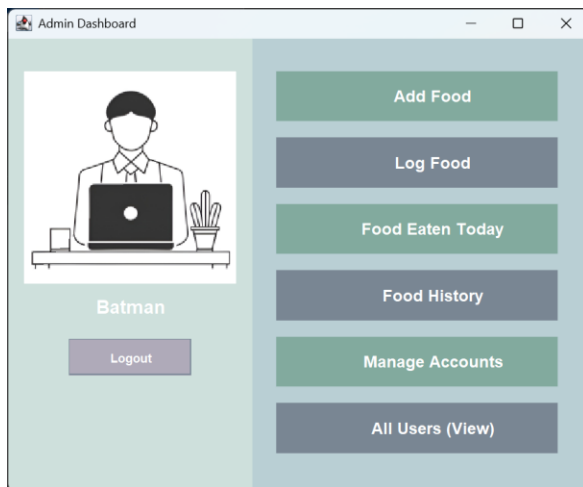
Nutrition Tracker - Login

NUTRITION TRACKER

Username

Password

Log In

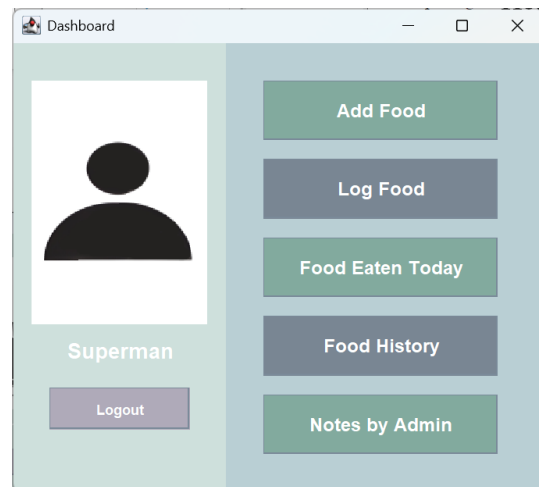


Admin Dashboard

Batman

Logout

- Add Food
- Log Food
- Food Eaten Today
- Food History
- Manage Accounts
- All Users (View)

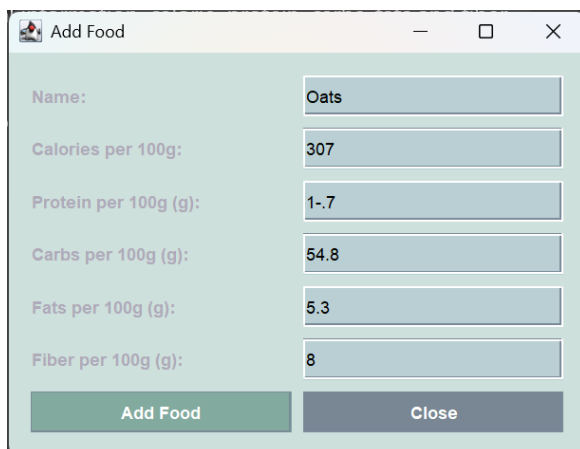


Dashboard

Superman

Logout

- Add Food
- Log Food
- Food Eaten Today
- Food History
- Notes by Admin



Add Food

Name:

Calories per 100g:

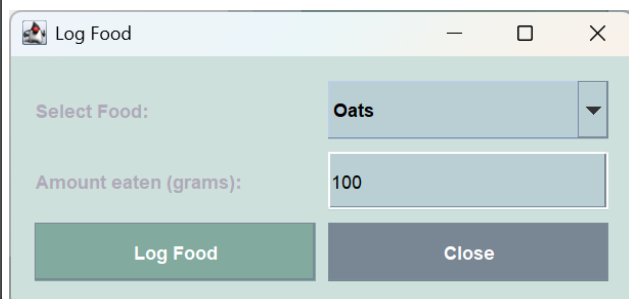
Protein per 100g (g):

Carbs per 100g (g):

Fats per 100g (g):

Fiber per 100g (g):

Add Food Close



Log Food

Select Food:

Amount eaten (grams):

Log Food Close

Food Eaten Today

Food	Grams	Calories	Protein	Carbs	Fats	Fiber
Oats	100.0	307.0	1.7	54.8	5.3	8.0

Totals: 307 cal, P1.7 C54.8 F5.3 Fib8.0

Food History

Date: 2025-08-22Show

Food	Grams	Calories	Protein	Carbs	Fats	Fiber
sat	200.0	40.0	40.0	40.0	40.0	40.0

Totals: 40 cal, P40.0 C40.0 F40.0 Fib40.0

Notes

Date	Note	From
2025-08-24T01:19:17	Increase fiber intake by 2grams	Batman

Manage Accounts

Create Account

Username Password RoleuserCreate

Change Password

User ID New PasswordChange Password

Delete Account

User IDDelete User

ID	Username	Role
1	Batman	admin
2	user	user
3	Superman	user
4	Ironman	admin
5	Ram Dahal	user

All Users

ID	Username	View
2	user	View
3	Superman	View
5	Ram Dahal	View

User Food & Notes

Date: 2025-08-23Show

Food	Grams	Calories	Protein	Carbs	Fats	Fiber
Oats	100.0	307.0	1.7	54.8	5.3	8.0

Totals: 307 cal, P1.7 C54.8 F5.3 Fib8.0

Add Note

Add Note

6. Conclusion

The Nutrition Tracker is a digital app that allows for digital diet and nutrient tracking with user friendly and straightforward summaries of activity daily. The Nutrition Tracker also minimizes manual errors and gives administrators a user interface for monitoring a user's actions. The Nutrition Tracker architecture is a modular construction having a DAO, a normalized schema, and a Swing UI easily maintains as well as easily accommodates enhancements of the health and nutrition trackers, and yet despite the Nutrition Tracker system being separable into parts, it provides a practical value to users for allowing them to make informed decisions on food and tracking, and in accordance with their unique health goals.

The administration or nutrition coach will benefit from having a way to influence user actions and offer help that leads to increased collaboration and accountability for their clients. This work has served in both constructing an applied example of software design principles and rules such as separation of concerns, role-based access, and relational model database design, while also demonstrating how an innovation in technology can be used to facilitate a management of lifestyle.

As a concrete application, the Nutrition Tracker serves as a functioning operational system for everyday use and also serves as a vehicle for the dietary and nutritional enhancements that transpire in the management of personal health and nutrition. It now exists as a solid base for further new and innovative uses of personal health and nutrition tracking.

7. Future Implementation

While the existing Nutrition Tracker has important functions with food entries as well as nutrient monitoring aspects, there are many things you could do down the road that would improve its strength, scaling ability, and usability.

1. Mobile app (Android/iOS) with shared API.
2. Analytics & dashboards (weekly trends, goal adherence, charts).
3. Cloud sync & multi-branch support (hosted DB, backups).
4. Food database import (CSV/USDA sources) and barcode scanning.
5. Notifications (daily reminder to log meals).
6. Coach workflows (plans, assignments, review queues)

Through these improvements, the system can further develop as a fully-fledged and intelligent nutrition management system that services current requirements and adapts to future technology changes and user expectations.